Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.1 – Newton's First and Second Law — A Crash and a Skydiver

Unit 2: Forces and Motion

Part 1 — Newton's First Law

Before you watch

You are about to watch part of a crash-test video presented by Dr. Griff Jones, showing what happens to a car — and the dummy inside it — when it is driven into a wall on purpose.

Listen for one word that comes up again and again: **inertia**.

While you watch

Fill these in as you go — you do not need full sentences yet.

1. What word does the video use for "the property of matter that resists a change in motion"?

   _____

2. A dummy is thrown forward in the demonstration with the tablecloth and the dishes. What does *that* demonstration show about objects that are **at rest**?

   _____

Newton's First Law

> **An object at rest will stay at rest, and an object in motion will stay in motion, unless acted upon by a net external force.**

Check: both halves say the same thing — nothing about an object's motion changes unless something from outside pushes or pulls on it.

Balanced and unbalanced forces

Every object has forces acting on it all the time. What matters is whether those forces are **balanced** or **unbalanced**:

Forces are…	**…so the object**
balanced	keeps doing whatever it was already doing — staying still, or moving at a steady speed in a straight line.
unbalanced	changes its motion — it speeds up, slows down, or changes direction.

In the crash, the wall applies an unbalanced force to the car, so the car's motion changes very suddenly. The dummy only changes its motion once something — the seatbelt, or the steering wheel — applies an unbalanced force to *it* too.

Apply it

1) **Explain** why a passenger still needs a seatbelt, even though the car itself has already stopped.

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Name: _____ **Date:** _____

Part 2 — Newton's Second Law

Before you watch

A skydiver jumps from a plane. Her speed changes several times before she lands — speeding up, then steady, then slowing down, then steady again. This video is only about three and a half minutes long. Watch it once all the way through, then we go back over it together.

Listen for one pair of words: **balanced** and **unbalanced**.

While you watch

1. Name the **two** forces acting on the skydiver as she falls.

   _____

2. One of those two forces stays the same the whole time. Which one?

   _____

3. The other force changes as she falls. Does it get bigger or smaller as her speed increases?

   _____

Newton's Second Law

> **When the forces on an object are unbalanced, the object accelerates — it speeds up, slows down, or changes direction — in the direction of the *bigger* force. When the forces are balanced, there is no acceleration: the object's speed stays constant.**

Check: "balanced" does not mean "stopped" — a skydiver at a steady speed has balanced forces and is still moving.

The skydive, in four stages

Fill in the last column yourself, using **speeding up**, **constant speed**, or **slowing down**.

Stage	Forces	What happens to her speed
Just after jumping	gravity much bigger than air resistance	
Falling for a while	air resistance has grown until it equals gravity	
Parachute opens	air resistance suddenly much bigger than gravity	
After the parachute has been open a while	air resistance has shrunk back down until it equals gravity again	

Apply it

1) **Explain** why the skydiver's falling speed stops increasing even though gravity has not changed.

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